Poornachandra G.

Lead Bioinformatics Scientist | Precision Medicine, Rare Diseases & Agentic AI Systems

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PROFESSIONAL SUMMARY

Bioinformatics Lead Scientist with 10+ years of experience in precision medicine across oncology and rare diseases. Expert in designing, building, and productionizing end-to-end applied bioinformatics and multi-omics pipelines for clinical workflows. Conceived and built comprehensive cross-functional bioinformatics stacks from scratch, delivering reproducible, scalable, and regulatory-ready solutions that integrate NGS analysis, clinical variant interpretation, phenotype-based prioritization, and data-driven insights. Active builder of AI-powered tools spanning multi-agent systems, clinical AI apps, and developer infrastructure.

SELECTED HIGHLIGHTS & IMPACT

- **Clinical NGS Pipelines:** Built and productionized end-to-end NGS pipelines (WGS/WES/TES, RNA-Seq, CNV, SV) used in clinical and research workflows; significantly reduced sample turnaround time through automated QC and reporting.
- **Variant Interpretation:** Architected variant annotation and prioritization workflows (ACMG/AMP/ASCO/CAP) integrating multiple public and proprietary annotations to enable actionable clinical decision support.
- **Biomarker Discovery & ML:** Led ML initiatives for biomarker discovery and patient stratification: feature engineering from multi-omics inputs, model validation, and deployment in production environments.
- **Rare Disease & Variant Selection:** Architected and built a tertiary analysis pipeline for phenotype-based gene prioritization, variant selection, and reporting in rare diseases. Presented this work and secured the top rank at the Bangalore Tech Summit (2025) as an independent researcher.
- **Team Leadership:** Mentored and grew a bioinformatics team; led technical hiring, performance reviews, and cross-functional training.
- **AI/LLM Engineering:** Built multiple AI projects spanning LLM tooling, multi-agent frameworks, clinical AI apps, and developer infrastructure. Established AI evaluation and observability frameworks (Promptfoo, DeepEval, Ragas) for accuracy benchmarking, hallucination analysis, and retrieval quality validation.

SKILLS & TECHNOLOGIES

Applied Bioinformatics & Clinical-omics	Rare Disease Genomics • Phenotype-based Gene Prioritization • WGS • WES • TES • RNA-Seq • scRNA-Seq • Liquid Biopsy • Companion Diagnostics • Variant Interpretation (ACMG/AMP/ASCO/CAP) • MSI • HRD • TMB • CNV • SV
Pipeline Engineering	GATK • Sentieon • DRAGEN • NVIDIA Parabricks • FreeBayes • DeepVariant • DeepSomatic • PureCN • CNVKit • VEP • ANNOVAR • Nextflow • QC Automation • Reproducibility • Kallisto • Salmon • DESeq2 • edgeR • Seurat • Scanpy • GSEA

Machine Learning & AI	Supervised/Unsupervised Learning • Feature Engineering • LightGBM • XGBoost • PyTorch • TensorFlow • scikit-learn • PyCaret • AutoML • PySpark • SHAP • scipy • pandas • NumPy • R
LLMs & Agentic Systems	LangChain • LangGraph • RAG Pipelines • Multi-Agent Systems • Prompt Engineering • Fine-Tuning • Quantization • AI Evaluation & Observability (Promptfoo, DeepEval, Ragas, LangSmith, Langfuse) • AI Validation (LLM-as-a-Judge, Red Teaming, Hallucination Analysis) • LiteLLM • MCP • W&B
Programming & Engineering	Python • Bash • R • FastAPI • SQLite • REST APIs
Cloud, Platforms & Infrastructure	AWS (EC2, S3, Batch, HealthOmics) • DNAnexus • BaseSpace • Galaxy • GenePattern • Terra/FireCloud • Illumina Connected Analytics • Docker • Git • CI/CD • Conda/Mamba • Jupyter • HPC • MLOps

PROFESSIONAL EXPERIENCE

Cellworks - Lead Scientist Bioinformatics

2024 - Present

- Led the Bioinformatics R&D team - defined the technical roadmap, recruited and mentored scientists, and drove project prioritization and delivery.
- Designed multi-agent orchestration workflows for vendor order management, coordinating specialized agents, external tools, and contextual memory with human-in-the-loop validation at critical decision points.
- Designed structured upskilling sessions leveraging supplemental projects beyond core deliverables, improving engagement, critical thinking, and problem-solving.
- Built multi-stage annotation and prioritization pipelines applying AMP/ASCO/CAP guidelines across public and proprietary annotation sources.
- Designed and maintained golden datasets, benchmark queries, and expected-answer sets that powered repeatable evaluation of retrieval quality, response accuracy, hallucination reduction, and regression testing across GenAI applications, achieving a 30% reduction in post-deployment defects.
- Built reusable prompt libraries and test suites (zero-shot, few-shot, role-based, multi-turn, edge-case) using Promptfoo and DeepEval to validate GenAI feature accuracy and consistency before release.

Cellworks - Senior Scientist, Reporting Manager

2021 - 2024

- Developed workflows for MSI, HRD, mutation signatures, TMB, and indel burden.
- Built high-throughput variant calling stacks on AWS following GATK best practices, including mutation and CNV calling, and biomarker extraction.
- Built and led the bioinformatics team: recruited, trained, mentored, and reviewed the work; created custom training protocols that accelerated onboarding and competency development.
- Designed input-agnostic workflows supporting multiple formats (FASTQ, BAM, VCF, structured reports) across diverse vendors.
- Built cancer driver and CNV vs. mRNA databases for 30+ cancer types using public datasets/tools.
- Developed automated multi-omics interpretation pipelines for high-impact clinical trial datasets from Takeda and Genentech.
- Created an in-house VUS reclassification pipeline using an Ensembl-based approach, improving classification accuracy by >40%.

Cellworks - Senior Scientist, Bioinformatics *2018 - 2021*

- Developed and benchmarked secondary analysis pipelines (Illumina DRAGEN, NVIDIA Parabricks, Sentieon), optimized for accuracy and turnaround time.
- Led custom panel design for liquid-biopsy samples; collaborated with Cleveland Clinic Florida to build an ML pipeline using longitudinal genomic data to predict early relapse in Melanoma.
- Built a framework to process pediatric cancer cases; served as primary owner for pediatric hematological cancers.
- Designed and implemented a Panel of Normals (PoN) workflow for CNV and mutation calling using ~100 cases, improving support for tumor-only analyses.
- Created multiple tertiary analysis workflows (GSEA, pathway enrichment, PCA, statistical models, ML pipelines) for biomarker discovery and treatment-response characterization.
- Developed multi-omics interpretation pipelines for the Foundation Medicine dataset (18,000+ cases).

Cellworks - Scientist, Clinical Translation *2017 - 2018*

- Curated rationale libraries to support therapy predictions, reducing turnaround time for live case analyses by >40%.
- Analyzed 200+ cancer cases, gaining deep expertise in target selection using cancer and drug networks in precision oncology.
- Conducted benchmarking of sequencing pipelines to establish confidence and reproducibility.
- Developed an in-house variant annotation and filtering pipeline, lowering processing costs and enabling greater customization.

Cellworks - Associate Scientist *2015 - 2017*

- Processed, interpreted, and harmonized multi-omics outputs for >500 cases, translating insights into automated data-harmonization and ETL pipelines.
- Curated and automated retrospective in-house dataset creation by integrating patient and cell-line data from >10,000 GDSC, TCGA, and published studies.
- Standardized genomic data interpretation from external partners.
- Automated report-generation workflows using reproducible pipelines and templates, reducing preparation time from ~15 hours to just minutes per week.

INDEPENDENT RESEARCH & RARE DISEASE COLLABORATIONS

- **Undiagnosed Diseases Network International (UDNI):** Active member and participant as an independent researcher, collaborating globally on rare disease research and contributing to the community of rare disease diagnostics.
- **Bangalore Tech Summit 2025 (Top Rank Winner):** Independently developed a comprehensive tertiary analysis pipeline for phenotype-based gene prioritization, clinical variant selection, and automated reporting. Presented the methodology and impact at the summit, securing the top rank.

EDUCATION

B.Tech, Biotechnology - JNTUA CEP (2015)

SELECTED PUBLICATIONS & ABSTRACTS

Co-author on 9 peer-reviewed publications and conference abstracts (J Neurooncol, J Clin Oncol, Blood/ASCO) on computational biomarker modeling and therapy-response prediction, 2018-2021. (Available on ORCID)